

Christopher Zarraga

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Computer Science student studying Artificial Intelligence and Systems Software in C, C++, and Python. Researching frame-monitoring of Autonomous Vehicles at UCSC’s AIEA Lab. Seeking 2027 Summer/Spring Internship!

EDUCATION

University of California - Santa Cruz

Expected June 2028

B.S. Computer Science (Applied Mathematics Minor)

Honors: **Full Ride Scholarship** (Sabatte Family 1st Cohort), Co-Founder Sabatte & Regents UCSC Association

Coursework: Artificial Intelligence, Parallel and Concurrent Programming, Systems Design, Database Systems

TECHNICAL SKILLS

Languages: C, **C++**, **Python**, TypeScript, JavaScript, SQL

Technologies: Linux, Bash, CMake, Git, Docker, Kubernetes, PostgreSQL, NumPy, PyTorch, Gymnasium, Stable-Baselines3

AI: Claude Code, Google Antigravity, Cursor, NVIDIA NeMo Agents, Ollama, OpenAI API

PROJECTS

Multi-Threaded HTTP Server | C, POSIX | [Kurisuo/multithreaded-http-server](#)

April 2026 – June 2026

- Built a web server in C sustaining **8400 requests per second at 64 simultaneous connections**, with 99% of requests completing in about 13ms on a 6-core VM
- Achieved a **2.2x speedup** by using a fixed-sized thread pool and bounded producer-consumer queue; implemented queue synchronization and per-URI reader-writer locks for concurrent reads, serialized writes, and audit log

Neural Network Inference Engine | C++17, Python, PyTorch | [Kurisuo/neural-inference-engine](#)

April 2026 – May 2026

- Built a C++17 MLP inference engine with no outside libraries, matching PyTorch’s predicted labels on all 10,000 MNIST test images at **24,000 images/second** on one CPU thread; loaded using a custom trained Pytorch Network
- Corrected a false-positive which compared results against a hardcoded accuracy causing an inaccuracy of 39 images; fixed it by computing the expected result at runtime and planting bugs to verify the tests catch the mistakes

VeriFi | C++17, nlohmann/JSON | [Kurisuo/VeriFi](#)

June 2026 – July 2026

- Founded and led a 5-person team** building a RAG application used to answer client questions using verified policy-based company documents; defined chunk formats, search API, and response schemas for parallel development
- Implemented and benchmarked a C++ cosine-similarity top-k search engine, matching most relevant passages at 0.36ms per search over 1k passages, scaling linearly to **96ms at 250k with 33k chunks/second ingestion**

NVPilot (NVIDIA x ASUS Hackathon) | TypeScript, NVML, NVIDIA NeMo | [Kurisuo/NVPilot](#)

May 2026 – July 2026

- Developed an AI Agent that utilizes NVML telemetry to tune system settings; achieved 550ms system perception, **sub-1ms rule-based planning**, and freed 2 GB of System RAM across 400+ background processes on an RTX 3060 laptop
- Made every change safe to undo: settings are checked against allowed ranges, each change logs how to reverse itself, and a fixed rule set takes over if the AI Agent fails; **rollback restores the original setup byte-for-byte in ~3s**

EXPERIENCE

AIEA Lab | Autonomous Vehicle Researcher (Frame-Monitoring)

May 2026 - Present

- Evaluated agent performance (Soft Actor Critic) in Gymnasium CarRacing using Stable-Baselines3, **analyzing reward curves and control telemetry** including steering, throttle, velocity, and acceleration
- Ran CARLA and reinforcement-learning workloads on the **Nautilus Kubernetes GPU cluster** using YAML-configured pods and Selkies Remote Desktop for autonomous-driving experiments

Perspective Paint | Freelance Web Developer

June 2026 – May 2026

- Rebuilt client’s website using HTML/CSS/JavaScript, implementing caching, asset compression, and optimized metadata; Google Analytics recorded **45% traffic growth over three months**; Secured client through cold outreach

Interests: Chess (**2000 Elo Peak**), Investing & Financial Literacy, Pickup Soccer (Goalie), Lap Swimming, Calisthenics

Events: SHPE 2026 Indianapolis, **Y-Combinator Intern Expo 2026**, NVIDIA x ASUS Hackathon, AMD x RedHat AI-Infra Workshop